

Discrete Mathematics

CHAPTER 4

Relations

Neso Academy



Introduction to Relations

Definition: Let A and B be two sets. A binary relation R from A to B is a subset of $A \times B$.

OR

$$R \subseteq A \times B$$

Recall: $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$

Usually we use the notation aRb to denote $(a, b) \in R$.

$a \not R b$ is used to denote $(a, b) \notin R$

Example: Let $A = \{1, 2, 3\}$ and $B = \{0, 1, 2, 4\}$
 $A \times B = \{(1, 0), (1, 1), (1, 2), (1, 4), (2, 0), (2, 1), (2, 2), (2, 4), (3, 0), (3, 1), (3, 2), (3, 4)\}$

Let say R is the relation where $(a, b) \in R$ if and only if $a = b$ then
 $R = \{(1, 1), (2, 2)\}$ and $R \subseteq A \times B$

Graphical representation of ordered pairs.



Arrows are used to represent ordered pairs of relation R .

Relation from a set to itself

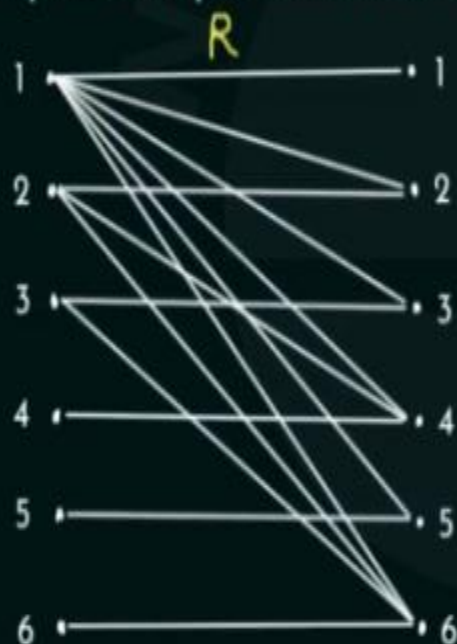
A relation on a set A is a relation from A to A .

Example: Let $R = \{(a, b) \mid a \text{ divides } b\}$

$$A = \{1, 2, 3, 4, 5, 6\}$$

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$

Graphical representation of relation R



Number of relations on a set with n elements

A relation on a set A is a subset of $A \times A$. Set A contains n elements and $A \times A$ contains n^2 elements

We are interested in listing down all the subsets of $A \times A$ means we are interested in finding the power set of $A \times A$.

We know that $P(A)$ is the power set of set A and let say A has n elements then $P(A)$ must have 2^n elements.

Therefore, $P(A \times A)$ must have 2^{n^2} elements.

Hence, number of relations on a set A with n elements $= |P(A \times A)| = 2^{n^2}$

Types of Relations (Part 1)

1. Reflexive Relation:

A relation R on a set A is called reflexive if $(a, a) \in R$ for every element $a \in A$. In other words, $\forall a((a, a) \in R)$.

Example: Let $A = \{1, 2, 3, 4\}$

$$R_1 = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (4, 4)\}$$

Relation R_1 is reflexive because it contains all ordered pairs of the form (a, a) for every element $a \in A$ i.e., R_1 has $(1, 1), (2, 2), (3, 3), (4, 4)$

$$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (4, 4)\}$$

Relation R_2 is not reflexive because the ordered pair $(3, 3)$ is not in R_2 .

2. Irreflexive Relation:

A relation R on a set A is called irreflexive if $\forall a \in A, (a, a) \notin R$.

Example: $A = \{1, 2, 3, 4\}$

$R_3 = \{(1, 2), (2, 1), (3, 3), (4, 4)\}$ is not irreflexive because $(3, 3)$ and $(4, 4)$ is there in R_3 .

$R_4 = \{(1, 2), (2, 1)\}$ is irreflexive because $\forall a \in A, (a, a) \notin R_4$

3. Symmetric Relation:

A relation R on a set A is called symmetric if $(b, a) \in R$ holds when $(a, b) \in R$ for all $a, b \in A$

In other words, relation R on a set A is symmetric if $\forall a \forall b ((a, b) \in R \rightarrow (b, a) \in R)$

Example: Relation $R_5 = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$ is symmetric because for every $(a, b) \in R_5$, $(b, a) \in R_5$ like $(1, 2)$ $(2, 1)$ is in R_5 .
There is no need to check for $(1, 1)$, $(2, 2)$.

Relation $R_6 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$ is not symmetric because for $(1, 2)$ there is no $(2, 1)$ in R_6 . Same is true for $(1, 3)$ and $(1, 4)$.

4. Antisymmetric Relation:

A relation R on a set A is called antisymmetric if $\forall a \forall b ((a, b) \in R \wedge (b, a) \in R \rightarrow (a = b))$
Whenever we have (a, b) in R , we will never have (b, a) in R until or unless $(a = b)$

Example: Relation $R_7 = \{(1, 1), (2, 1)\}$ on set A is antisymmetric because $(2, 1)$ is in R_7 but $(1, 2)$ is not in R_7 .

REFLEXIVE RELATION:

Let R be a relation on a set A . R is reflexive if and only if, for all $a \in A$, $(a, a) \in R$ or equivalently aRa . That is, each element of A is related to itself.

REMARK

R is not reflexive iff there is an element “ a ” in A such that $(a, a) \notin R$. That is, some element “ a ” of A is not related to itself.

EXAMPLE:

Let $A = \{1, 2, 3, 4\}$ and define relations R_1, R_2, R_3, R_4 on A as follows:

$$R_1 = \{(1, 1), (3, 3), (2, 2), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 3)\}$$

$$R_3 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_4 = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 4)\}$$

Then,

R_1 is reflexive, since $(a, a) \in R_1$ for all $a \in A$.

R_2 is not reflexive, because $(4, 4) \notin R_2$.

R_3 is reflexive, since $(a, a) \in R_3$ for all $a \in A$.

R_4 is not reflexive, because $(1, 1) \notin R_4, (3, 3) \notin R_4$

Matrix representation of Reflexive Relation

	1	2	3	4
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1

	1	2	3	4
1	1	0	0	1
2	0	1	0	0
3	0	0	1	0
4	0	0	1	0

	1	2	3	4
1	1	1	0	0
2	1	1	0	0
3	0	0	1	0
4	0	0	0	1

	1	2	3	4
1	0	0	1	0
2	0	1	0	1
3	1	0	0	0
4	0	0	0	1

Matrix representation of Irreflexive Relation

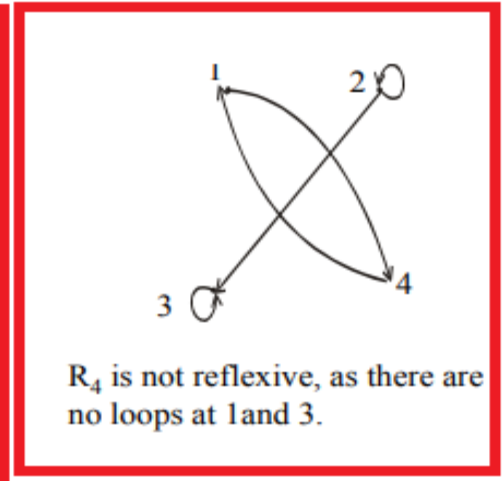
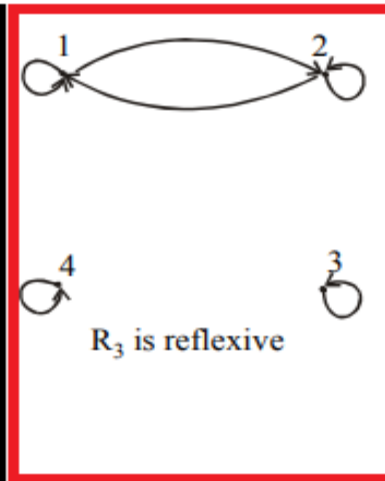
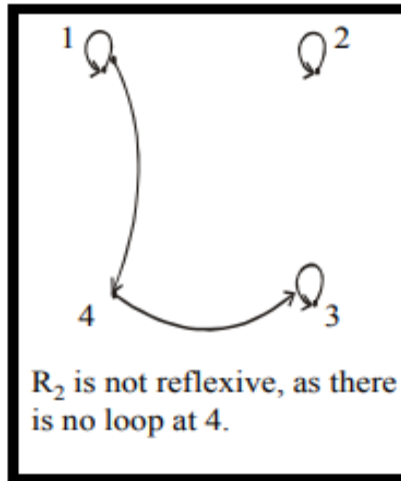
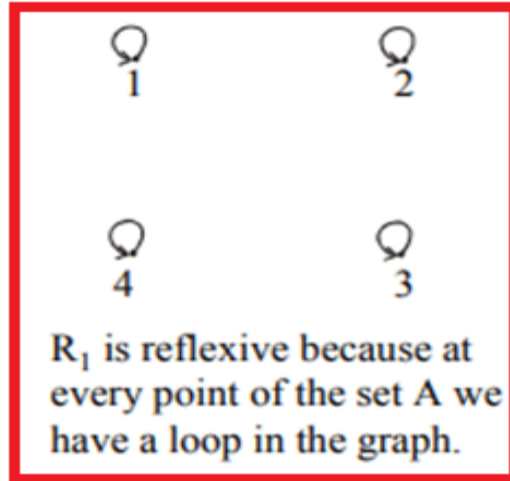
	1	2	3	4
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0

	1	2	3	4
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0

	1	2	3	4
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0

	1	2	3	4
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0

Loop representation of Reflexive Relation



Loop representation of Irreflexive Relation

NO LOOP INVOLVES

SYMMETRIC RELATION

Let R be a relation on a set A . R is symmetric if, and only if, for all $a, b \in A$, if $(a, b) \in R$, then $(b, a) \in R$.

That is, if aRb then bRa .

EXAMPLE

Let $A = \{1, 2, 3, 4\}$ and define relations R_1, R_2, R_3 , and R_4 on A as follows.

$$R_1 = \{(1, 1), (1, 3), (2, 4), (3, 1), (4, 2)\}$$

$$R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_3 = \{(2, 2), (2, 3), (3, 4)\}$$

$$R_4 = \{(1, 1), (2, 2), (3, 3), (4, 3), (4, 4)\}$$

Then R_1 is symmetric because for every order pair (a, b) in R_1 also have (b, a) in R_1 . For example, we have $(1, 3)$ in R_1 then we have $(3, 1)$ in R_1 . Similarly all other ordered pairs can be checked.

R_2 is also symmetric. We say it is vacuously true.

R_3 is not symmetric, because $(2, 3) \in R_3$ but $(3, 2) \notin R_3$.

R_4 is not symmetric because $(4, 3) \in R_4$ but $(3, 4) \notin R_4$.

Matrix representation of Symmetric Relation

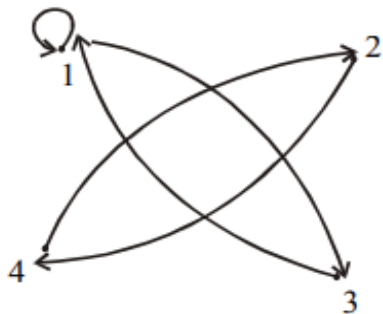
	1	2	3	4
1	1	0	1	0
2	0	0	0	1
3	1	0	0	0
4	0	1	0	0

	1	2	3	4
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1

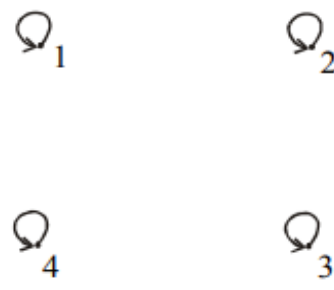
	1	2	3	4
1	0	0	0	0
2	0	1	1	0
3	0	0	0	1
4	0	0	0	0

	1	2	3	4
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	1	1

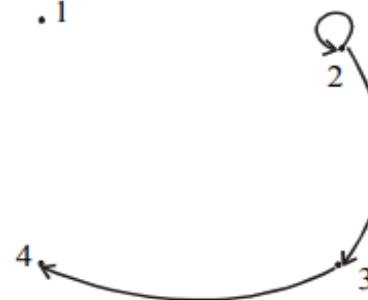
Loop representation of Symmetric Relation



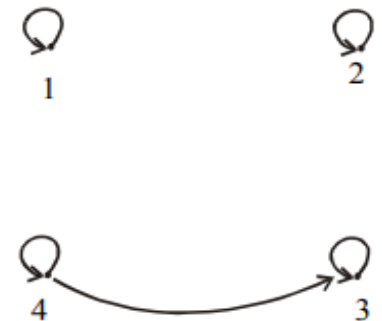
R_1 is symmetric



R_2 is symmetric



R_3 is not symmetric since there are arrows from 2 to 3 and from 3 to 4 but not conversely



R_4 is not symmetric since there is an arrow from 4 to 3 but no arrow from 3 to 4

ASYMMETRIC RELATION

In [set theory](#), A relation R on a set A is called asymmetric if no $(y,x) \in R$ when $(x,y) \in R$. Or we can say, the relation R on a set A is asymmetric if and only if, $(x,y) \in R \implies (y,x) \notin R$.

A SYMMETRIC RELATION

A relation R is called asymmetric if $(a, b) \in R$ implies that $(b, a) \notin R$.

Self-symmetric
A Symmetric
A/S Symmetric

	1	2	3
1	1	1	1
2	1	1	0
3	1	0	0

$$R = \{(1,1), (1,2), (1,3), (2,1), (2,2)\}$$

$$R = \{(1,1), (1,2), (1,3), (2,1), (2,2)\}$$

NOT A SYMMETRIC

	1	2	3	4
1	0	1	1	0
2	0	0	0	0
3	1	0	0	0
4	0	0	1	0

$$R = \{(1,2), (1,3), (3,1), (4,3)\}$$

$$R = \{(1,2), (1,3), (3,1), (4,3)\}$$

NOT A SYMMETRIC

	1	2	3	4
1	0	1	0	0
2	0	0	1	0
3	1	0	0	0
4	0	0	1	0

$$R = \{(1,2), (2,3), (3,1), (4,3)\}$$

A SYMMETRIC

ANTISYMMETRIC RELATION:

Let R be a binary relation on a set A . R is **anti-symmetric** iff

$$\forall a, b \in A \text{ if } (a,b) \in R \text{ and } (b,a) \in R \text{ then } a = b.$$

REMARK:

- 1) R is not **anti-symmetric** iff there are elements a and b in A such that $(a, b) \in R$ and $(b, a) \in R$ but $a \neq b$.
- 2) The properties of being **symmetric** and being **anti-symmetric** are not negative of each other.

EXAMPLE:

Let $A = \{1,2,3,4\}$ and define the following relations on A .

$$R_1 = \{(1,1), (2,2), (3,3)\}$$

$$R_2 = \{(1,2), (2,2), (2,3), (3,4), (4,1)\}$$

$$R_3 = \{(1,3), (2,2), (2,4), (3,1), (4,2)\}$$

$$R_4 = \{(1,3), (2,4), (3,1), (4,3)\}$$

R_1 is anti-symmetric and symmetric.

R_2 is anti-symmetric but not symmetric because $(1,2) \in R_2$ but $(2,1) \notin R_2$.

R_3 is not anti-symmetric since $(1,3) \& (3,1) \in R_3$ but $1 \neq 3$.

Note that R_3 is symmetric.

R_4 is neither anti-symmetric because $(1,3) \& (3,1) \in R_4$ but $1 \neq 3$ nor symmetric because $(2,4) \in R_4$ but $(4,2) \notin R_4$

Matrix representation of Ant symmetric Relation

	1	2	3	4
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1

	1	2	3	4
1	0	1	0	0
2	0	1	1	0
3	0	0	0	1
4	1	0	0	0

	1	2	3	4
1	0	0	1	0
2	0	1	0	1
3	1	0	0	0
4	0	1	0	0

	1	2	3	4
1	0	0	1	0
2	0	0	0	1
3	1	0	0	0
4	0	0	1	0

TRANSITIVE RELATION

Let R be a relation on a set A . R is transitive if and only if for all $a, b, c \in A$, if $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$.

That is, if aRb and bRc then aRc .

In words, if any one element is related to a second and that second element is related to a third, then the first is related to the third.

Note: The “first”, “second” and “third” elements need not to be distinct.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$ and define relations R_1 , R_2 and R_3 on A as follows:

$$R_1 = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$$

$$R_2 = \{(1, 2), (1, 4), (2, 3), (3, 4)\}$$

$$R_3 = \{(2, 1), (2, 4), (2, 3), (3, 4)\}$$

Then R_1 is transitive because $(1, 1)$, $(1, 2)$ are in R , then to be transitive relation $(1, 2)$ must be there and it belongs to R .

Similarly for other order pairs. R_2 is not transitive since $(1, 2)$ and $(2, 3) \in R_2$ but $(1, 3) \notin R_2$.

R_3 is transitive.

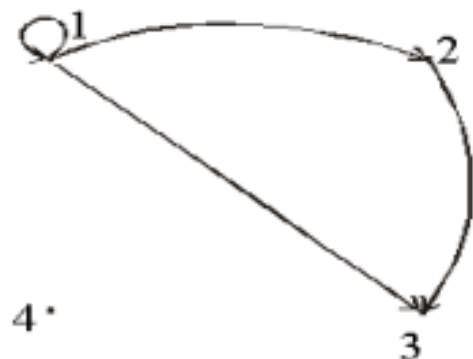
EXAMPLE

Let $A = \{1, 2, 3, 4\}$ and define relations R_1 , R_2 and R_3 on A by the directed graphs:

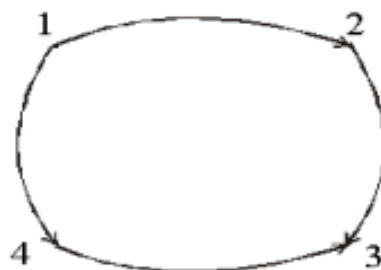
$$R_1 = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$$

$$R_2 = \{(1, 2), (1, 4), (2, 3), (3, 4)\}$$

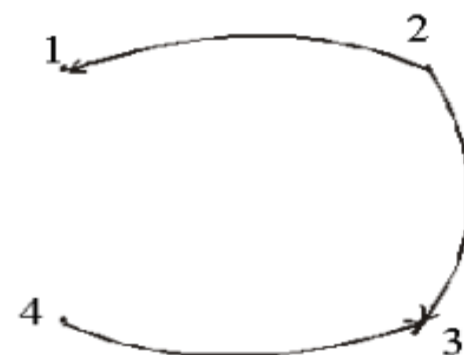
$$R_3 = \{(2, 1), (2, 4), (2, 3), (3, 4)\}$$



R_1 is transitive



R_2 is not transitive since there is an arrow from 1 to 2 and from 2 to 3 but no arrow from 1 to 3 directly



R_3 is transitive

EQUIVALANCE RELATION EXAMPLE NO 1

Reflexive				
	0	1	2	3
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1

Symmetric				
	0	1	2	3
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1

Transitive				
	0	1	2	3
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1

$$R = \{(0,0), (1,1), (2,2), (3,3)\}$$

EQUIVALANCE RELATION EXAMPLE NO 2

Reflexive				
	0	1	2	3
0	1	0	1	0
1	0	1	0	0
2	1	0	1	1
3	0	0	1	1

Symmetric				
	0	1	2	3
0	1	0	1	0
1	0	1	0	0
2	1	0	1	1
3	0	0	1	1

Transitive				
	0	1	2	3
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1

$$R = \{(0,0), (0,2), (2,0), (1,1), (2,2), (2,3), (3,2), (3,3)\}$$

NOT A EQUIVALANCE RELATION

Reflexive				
	0	1	2	3
0	1	0	1	0
1	0		0	0
2	1	0	1	1
3	0	0	1	1

Symmetric				
	0	1	2	3
0	1	0	1	0
1	0		0	0
2	1	0	1	1
3	0	0	1	1

Transitive				
	0	1	2	3
0	1	0	0	0
1	0		0	0
2	0	0	1	0
3	0	0	0	1

$$R = \{(0,0), (0,2), (2,0), (2,2), (2,3), (3,2), (3,3)\}$$

Types of Relations (Part II)

5. Transitive Relation:

A relation R on a set A is called transitive if $\forall a \forall b \forall c ((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R$

Example: $A = \{1, 2, 3, 4\}$

$R_8 = \{(2, 1), (3, 1), (3, 2), (4, 4)\}$ is transitive because $(3, 2)$, $(2, 1)$, and $(3, 1)$ are there in R_8 .

$R_9 = \{(2, 1), (1, 3)\}$ is not transitive as $(2, 1)$ and $(1, 3)$ are there in R_9 but there is no $(2, 3)$ in relation R_9 .

6. Asymmetric Relation:

A relation R on a set A is called asymmetric if $\forall a \forall b ((a, b) \in R \rightarrow (b, a) \notin R)$

Example: $A = \{1, 2, 3, 4\}$

$R_{10} = \{(1, 1), (1, 2), (1, 3)\}$ is not an asymmetric relation because of $(1, 1)$.

$R_{11} = \{(1, 2), (1, 3), (2, 3)\}$ is an asymmetric relation.

Summary

Relation

1. Reflexive
2. Irreflexive
3. Symmetric
4. Antisymmetric
5. Asymmetric
6. Transitive

Property

$$\forall a((a, a) \in R)$$

$$\forall a((a, a) \notin R)$$

$$\forall a \forall b((a, b) \in R \rightarrow (b, a) \in R)$$

$$\forall a \forall b(((a, b) \in R \wedge (b, a) \in R) \rightarrow (a = b))$$

$$\forall a \forall b((a, b) \in R \rightarrow (b, a) \notin R)$$

$$\forall a \forall b \forall c(((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R)$$

Types of Relations (Solved Problem)

Problem: Determine whether relation R on the set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

- a) $x + y = 0$ b) $x - y$ is a rational number c) $xy = 0$
d) $x = 1$ or $y = 1$

Types of Relations (Solved Problem)

Problem: Determine whether relation R on the set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

- a) $x + y = 0$ b) $x - y$ is a rational number c) $xy = 0$
d) $x = 1$ or $y = 1$

Solution: a) $R = \{(x, y) \mid x + y = 0\}$

1. Reflexive: $\forall a \in \mathbb{R} ((a, a) \in R)$
Not reflexive. Only true for $(0, 0)$.

2. Symmetric:
 $\forall a \forall b \in \mathbb{R} ((a, b) \in R \rightarrow (b, a) \in R)$
If $a + b = 0$ then $b + a = 0$.
Therefore, R is symmetric.

3. Antisymmetric:
 $\forall a \forall b \in \mathbb{R} ((a, b) \in R \wedge (b, a) \in R) \rightarrow (a = b)$
Not antisymmetric.
 $1 + (-1) = 0$ and $-1 + 1 = 0$ but $1 \neq -1$

4. Transitive:
 $\forall a \forall b \forall c ((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R$
Not Transitive. $1 + (-1) = 0$ and $(-1) + 1 = 0$
but $1 + 1 \neq 0$

b) $R = \{(x, y) \mid x-y \text{ is a rational no.}\}$

1. Reflexive: $a - a = 0$ is a rational number.
2. Symmetric:
if $a-b$ is a rational number
then $b-a = -(a-b)$ is also a

c) $R = \{(x, y) \mid xy = 0\}$

1. Reflexive: Not reflexive
 $a \cdot a = a^2 = 0$ when $a = 0$.
2. Symmetric:
 $ab = 0$ then $ba = 0$
3. Antisymmetric:
Not antisymmetric.
 $1 \times 0 = 0$ and $0 \times 1 = 0$ but $1 \neq 0$
4. Transitive: Not transitive
 $-1 \times 0 = 0$ and $0 \times 2 = 0$ but
 $-1 \times 2 \neq 0$.

rational number.

3. Antisymmetric: Not antisymmetric
 $3 - 2$ and $2 - 3$ are both rational numbers
but $3 \neq 2$.

4. Transitive:

If $a - b$ is a rational number and $b - c$
is also a rational number, then $a - c$
 $= (a-b) + (b-c)$ is also a rational number.

d) $R = \{(x, y) \mid x = 1 \text{ or } y = 1\}$

1. Reflexive: Not reflexive
 $(0, 0)$ is not in R .
2. Symmetric:
if $(a, b) \in R$ then $a = 1$ or $b = 1$.
This means $(b, a) \in R$.
3. Antisymmetric:
Not antisymmetric.
if $a = 1$ and $b = 0$ then $(a, b) \in R$ and
 $(b, a) \in R$ but $a \neq b$.
4. Transitive:

Homework

Types of Relations (GATE Problems)

Problem 1: Amongst the properties {reflexivity, symmetry, antisymmetry, transitivity}, the relation $R = \{(x, y) \in \mathbb{N}^2 \mid x \neq y\}$ satisfies?

[GATE 1994]

Solution:

1. Reflexivity: $\forall a((a, a) \in R)$
 $a \neq a$ (Not possible).
Therefore, R is not reflexive.
2. Symmetry: $\forall a \forall b((a, b) \in R \rightarrow (b, a) \in R)$
 $a \neq b$ then $b \neq a$
Therefore, R is symmetric.
3. Antisymmetry: $\forall a \forall b(((a, b) \in R \wedge (b, a) \in R) \rightarrow (a = b))$
 $(a \neq b \wedge b \neq a) \rightarrow (a = b)$ is false.
Therefore, R is not antisymmetric.
4. Transitivity: $\forall a \forall b \forall c(((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R)$
 $(a \neq b \wedge b \neq c) \rightarrow (a \neq c)$
Need not be true.
Let $a = 1$, $b = 2$, and $c = 1$
 $1 \neq 2$ and $2 \neq 1$ but $1 = 1$.
Therefore, R is not transitive.

Problem 2: The binary relation $S = \emptyset$ (empty set) on a set $A = \{1, 2, 3\}$ is

- (A) Neither reflexive nor symmetric.
- (B) Symmetric and reflexive.
- (C) Transitive and reflexive.
- (D) Transitive and symmetric.

[GATE 2002]

Solution: $S = \emptyset$

(i) Reflexivity: $\forall a \in A ((a, a) \in S)$

S is an empty set and there is not even a single pair (a, a) inside S .
Therefore, S is not reflexive.

(ii) Symmetry: $\forall a \forall b \in A ((a, b) \in S \rightarrow (b, a) \in S)$

$S = \emptyset$ is symmetric.

because $(a, b) \in S$ is false. Hence, implication is always true.
Therefore, empty set is symmetric.

(iii) Transitivity: $\forall a \forall b \forall c \in A (((a, b) \in S \wedge (b, c) \in S) \rightarrow (a, c) \in S)$

$S = \emptyset$ is transitive.

because $(a, b) \in S \wedge (b, c) \in S$ is false.

Hence, implication is always true.

Therefore, option (D) is the correct option.

Note: empty set is antisymmetric, irreflexive, and asymmetric.

Problem 2: The binary relation $S = \emptyset$ (empty set) on a set $A = \{1, 2, 3\}$ is

- (A) Neither reflexive nor symmetric.
- (B) Symmetric and reflexive.
- (C) Transitive and reflexive.
- ✓ (D) Transitive and symmetric.

[GATE 2002]

Types of Relations (GATE Problems)

Problem 1: Consider the binary relation $R = \{(x, y), (x, z), (z, x), (z, y)\}$ on the set $\{x, y, z\}$. Which one of the following is TRUE?

- (A) R is symmetric but NOT antisymmetric.
- (B) R is NOT symmetric but antisymmetric.
- (C) R is both symmetric and antisymmetric.
- (D) R is neither symmetric nor antisymmetric.

[GATE 2009]

Solution:

$$R = \{(x, y), (x, z), (z, x), (z, y)\}$$

(i) Symmetric: $\forall a \forall b ((a, b) \in R \rightarrow (b, a) \in R)$

$$(x, y) \in R \text{ but } (y, x) \notin R$$

Therefore, R is not symmetric.

(ii) Antisymmetric: $\forall a \forall b ((a, b) \in R \wedge (b, a) \in R \rightarrow (a=b))$

$$(x, z) \in R \wedge (z, x) \in R \text{ but } x \neq z$$

Therefore, R is not antisymmetric.

Problem 1: Consider the binary relation $R = \{(x, y), (x, z), (z, x), (z, y)\}$ on the set $\{x, y, z\}$. Which one of the following is TRUE?

- (A) R is symmetric but NOT antisymmetric.
- (B) R is NOT symmetric but antisymmetric.
- (C) R is both symmetric and antisymmetric.
- ✓(D) R is neither symmetric nor antisymmetric.

[GATE 2009]

Problem 2: Let R be a relation on the set of ordered pairs of positive integers such that $((p, q), (r, s)) \in R$ if and only if $p - s = q - r$. Which one of the following is TRUE about R ?

- (A) Both reflexive and symmetric.
- (B) Reflexive but not symmetric.
- (C) Not reflexive but symmetric.
- (D) Neither reflexive nor symmetric.

[GATE 2015]

Solution: R is a relation on a set of ordered pairs of positive integers.

Let $A = \{(1, 1), (1, 2), (1, 3), \dots\}$

R is a relation on a set A.

Therefore, $R = \{((p, q), (r, s)) \mid p - s = q - r\}$

1. Reflexive:

Original - $\forall a \in A ((a, a) \in R)$

New - $\forall (a, b) \in A ((a, b), (a, b)) \in R$

$R = \{((p, q), (r, s)) \mid p - s = q - r\}$

$a - b = b - a$ [Not possible for any positive integers a and b]

Therefore, R is not reflexive.

2. Symmetric:

Original - $\forall a \forall b \in A ((a, b) \in R \rightarrow (b, a) \in R)$

New - $\forall (a, b) \forall (c, d) \in A (((a, b), (c, d)) \in R \rightarrow ((c, d), (a, b)) \in R)$

$R = \{((p, q), (r, s)) \mid p - s = q - r\}$

$((a, b), (c, d)) \in R$ means $a - d = b - c$

$((c, d), (a, b)) \in R$ means $c - b = d - a$

$[a - d = b - c] \rightarrow [c - b = d - a]$

$$a - d = b - c$$

$$a - b - d = -c$$

$$a - b = -c + d$$

$$a - b + c = d$$

$$c - b = d - a$$

$\therefore R$ is symmetric

Problem 2: Let R be a relation on the set of ordered pairs of positive integers such that $((p, q), (r, s)) \in R$ if and only if $p - s = q - r$. Which one of the following is TRUE about R ?

- (A) Both reflexive and symmetric.
- (B) Reflexive but not symmetric.
- ✓ (C) Not reflexive but symmetric.
- (D) Neither reflexive nor symmetric.

[GATE 2015]

Operations on Relations

Let us consider two relations R_1 and R_2 from set A to B .

These relations can be combined in a way two sets can be combined. i.e., $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, $R_2 - R_1$, $R_1 \oplus R_2$.

Example: Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$

$$R_1 = \{(1, 3), (1, 4), (3, 3)\}$$

$$R_2 = \{(2, 3), (2, 4), (3, 3), (2, 5)\}$$

$$R_1 \cup R_2 = \{(1, 3), (1, 4), (2, 3), (2, 4), (2, 5), (3, 3)\}$$

$$R_1 \cap R_2 = \{(3, 3)\}$$

$$R_1 - R_2 = \{(1, 3), (1, 4)\}$$

$$R_2 - R_1 = \{(2, 3), (2, 4), (2, 5)\}$$

$$R_1 \oplus R_2 = (R_1 \cup R_2) - (R_1 \cap R_2) = \{(1, 3), (1, 4), (2, 3), (2, 4), (2, 5)\}$$

Problem: Suppose that R and S are reflexive relations on a set A . Prove or disprove each of these statements.

- a) $R \cup S$ is reflexive.
- b) $R \cap S$ is reflexive.
- c) $R \oplus S$ is irreflexive.
- d) $R - S$ is irreflexive.

Solution:

a) R is reflexive
 $\forall a \in A (a, a) \in R$
S is reflexive
 $\forall a \in A (a, a) \in S$

$R \cup S$ must contain all elements in either R or S.
Therefore, $R \cup S$ is reflexive.

b) R is reflexive
 $\forall a \in A (a, a) \in R$
S is reflexive
 $\forall a \in A (a, a) \in S$

$R \cap S$ must contain all elements in both R and S.
Therefore, $R \cap S$ is reflexive.

c) R is reflexive
 $\forall a \in A (a, a) \in R$
S is reflexive
 $\forall a \in A (a, a) \in S$

$R \oplus S$ contains all elements in R or S but not in both.
 $\forall a \in A (a, a) \notin R \oplus S$
Therefore, $R \oplus S$ is irreflexive.

d) R and S are reflexive.

R-S contains all elements in R that are not in S.

$\forall a \in A (a, a) \notin R-S$

Therefore, R - S is irreflexive.

Example: $A = \{1, 2, 3\}$

$R = \{(1, 1), (2, 2), (3, 3), (3, 4)\}$

$S = \{(1, 1), (2, 2), (3, 3), (2, 3)\}$

$R \cup S = \{(1, 1), (2, 2), (3, 3), (2, 3), (3, 4)\}$

$R \cap S = \{(1, 1), (2, 2), (3, 3)\}$

$R \oplus S = \{(3, 4), (2, 3)\}$

$R - S = \{(3, 4)\}$

Composition of Relations

Let A , B and C be three sets. Suppose, R is a relation from A to B , and S is a relation from B to C . The composite of R and S , denoted by SoR , is a binary relation from A to C consisting of ordered pairs (a, c) where $a \in A$ and $c \in C$. Also, $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$.

Example: What is the composite of the relations R and S , where R is the relation from set $A = \{1, 2, 3\}$ to set $B = \{3, 4, 5\}$ with $R = \{(1, 3), (1, 4), (1, 5), (2, 3), (3, 3)\}$ and S is the relation from set $B = \{3, 4, 5\}$ to set $C = \{1, 2, 3, 4\}$ with $S = \{(3, 1), (3, 2), (3, 3), (4, 3), (4, 4)\}$

Solution:



$$SoR = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

Note: If R and S are relations from A to B and C to D respectively, then SoR is not defined unless $B = C$.

Composition of Relation with itself

Problem: Let R be the relation on the set $\{1, 2, 3, 4, 5\}$ containing the ordered pairs $(1, 1)$, $(1, 2)$, $(1, 3)$, $(2, 3)$, $(2, 4)$, $(3, 1)$, $(3, 4)$, $(3, 5)$, $(4, 2)$, $(4, 5)$, $(5, 1)$, $(5, 2)$ and $(5, 4)$.

Find

- a) R^2 b) R^3 c) R^4

Solution: $R = \{(1, 1), (1, 2), (1, 3), (2, 3), (2, 4), (3, 1), (3, 4), (3, 5), (4, 2), (4, 5), (5, 1), (5, 2), (5, 4)\}$

$R = \{(1, 1), (1, 2), (1, 3), (2, 3), (2, 4), (3, 1), (3, 4), (3, 5), (4, 2), (4, 5), (5, 1), (5, 2), (5, 4)\}$

a) R^2 is the composite of R with itself and it consists of ordered pairs (a, c) such that $(a, b) \in R$ and $(b, c) \in R$.

$$R^2 = R \circ R$$

$R \circ R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 4), (2, 5), (2, 2), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (4, 1), (4, 2), (4, 3), (4, 4), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5)\}$

b) $R^3 = R^2 \circ R$

$= \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5)\}$

c) $R^4 = R^3 \circ R$

R^3 already contains all possible ordered pairs in $A \times A$, R^4 will also contain all possible ordered pairs.

Therefore, $R^4 = R^3$

Representation of Relations

Let say we have two sets A and B.

$$A = \{1, 2, 3\}$$

$$B = \{0, 1, 2, 4\}$$

There are multiple ways to represent a relation.

Let say we want to represent a relation R which consists of all ordered pairs (x, y) where $x \in A$ and $y \in B$ and $x \leq y$.

1. Listing Method: $R = \{(1, 1), (1, 2), (1, 4), (2, 2), (2, 4), (3, 4)\}$

2. Set Builder Method:

$$R_{A \text{ to } B} = \{(x, y) \mid x \leq y\}$$

OR

$$R = \{(x, y) \mid x \in A \wedge y \in B \wedge x \leq y\}$$

3. Statement Representation:

$$\forall x \in A \forall y \in B, xRy \text{ iff } x \leq y$$

4. Matrix Representation:

$$\begin{matrix} & \begin{matrix} 0 & 1 & 2 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}_{m \times n}$$

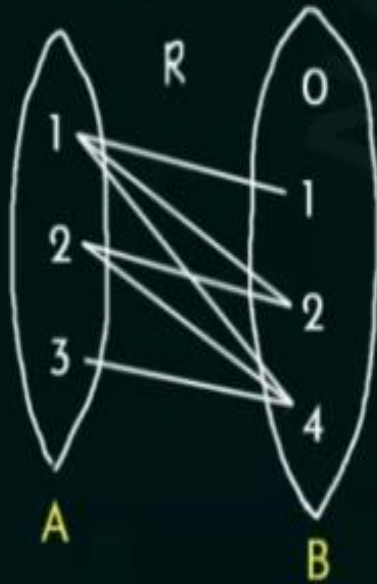
$$\begin{aligned} |A| &= m \\ |B| &= n \end{aligned}$$

5 Graph Representation:

Directed Graph



6. Arrow Diagram Representation:



Closure of Relations (Part 1)

Reflexive Closure:

Let say we have a binary relation R.

$R = \{(1, 1), (2, 2), (2, 3)\}$ defined on a set $A = \{1, 2, 3\}$.

The relation R is not reflexive.

Smallest reflexive relation that contains R must include the ordered pair (3, 3).

$$R_{\text{New}} = \{(1, 1), (2, 2), (3, 3), (2, 3)\}$$

Definition: Reflexive closure of a binary relation R on a set A is the smallest reflexive relation of the set A that contains R.

Reflexive closure of R is usually denoted by R_r^+ .

$$R_r^+ = R \cup \{(a, a) \mid a \in A\}$$

Problem: Let R be the relation on the set $\{0, 1, 2, 3\}$ containing the ordered pairs $(0, 1)$, $(1, 1)$, $(1, 2)$, $(2, 0)$, $(2, 2)$, and $(3, 0)$. Find the reflexive closure of R .

Solution: $R = \{(0, 1), (1, 1), (1, 2), (2, 0), (2, 2), (3, 0)\}$
 $A = \{0, 1, 2, 3\}$

Reflexive closure of R

$$R_r^+ = R \cup \{(a, a) \mid a \in A\} \quad R_r^+ = \{(0, 1), (1, 1), (1, 2), (2, 0), (2, 2), (3, 0), (0, 0), (3, 3)\}$$

This relation

1. contains R .
2. is reflexive.
3. is minimal.

Closure of Relations (Part 2)

Symmetric closure: Symmetric closure of a binary relation R on a set A is the smallest symmetric relation on a set A that contains R .

$$R_s^+ = R \cup \{(b, a) \mid (a, b) \in R\}$$

Problem: Let R be the relation on the set $\{0, 1, 2, 3\}$ containing the ordered pairs $(0, 1)$, $(1, 1)$, $(1, 2)$, $(2, 0)$, $(2, 2)$, and $(3, 0)$. Find the symmetric closure of R .

Solution: $R = \{(0, 1), (1, 1), (1, 2), (2, 0), (2, 2), (3, 0)\}$
 $A = \{0, 1, 2, 3\}$



$$R_s^+ = R \cup \{(b, a) \mid (a, b) \in R\}$$

$$R_s^+ = \{(0, 1), (1, 1), (1, 2), (2, 0), (2, 2), (3, 0), (1, 0), (2, 1), (0, 2), (0, 3)\}$$

$$R_s^+ = \{(0, 1), (0, 2), (0, 3), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2), (3, 0)\}$$

Transitive closure: Transitive closure of a binary relation R on a set A is the smallest transitive relation on a set A that contains R .

$$R_{\dagger}^{+} = R \cup \{(a, c) \mid (a, b) \in R \wedge (b, c) \in R\}$$

Problem: Let R be the relation on a set $\{1, 2, 3\}$ containing the ordered pairs $(1, 1)$, $(2, 3)$, and $(3, 1)$. Find the transitive closure of R .

Solution: $R = \{(1, 1), (2, 3), (3, 1)\}$
 $A = \{1, 2, 3\}$

$$R_{\dagger}^{+} = R \cup \{(a, c) \mid (a, b) \in R \wedge (b, c) \in R\}$$

$$R_{\dagger}^{+} = \{(1, 1), (2, 3), (3, 1), (2, 1)\}$$

Closure of Relations (Solved Problems)

Problem 1: Let R be the relation $\{(a, b) \mid a \neq b\}$ on the set of integers. What is the reflexive closure of R ?

Solution: $R = \{(a, b) \mid a \neq b\}$
Relation R is defined on set of integers.

$$R_r^+ = R \cup \{(a, a) \mid a \in A\} \quad [A \text{ is the set of all integers}]$$
$$= \{(a, b) \mid a \neq b\} \cup \{(a, b) \mid a = b\}$$

This means all pairs of integers must be included in the reflexive closure of R .

$$= \{(a, b) \mid a, b \in \mathbb{Z}\}$$
$$= \mathbb{Z} \times \mathbb{Z}$$

Therefore, $R_r^+ = \mathbb{Z} \times \mathbb{Z}$

Problem 2: Let R be the relation $\{(a, b) \mid a \text{ divides } b\}$ on the set of integers. What is the symmetric closure of R ?

Solution: $R = \{(a, b) \mid a \text{ divides } b\}$
 $A = \text{set of integers} = \mathbb{Z}$

$$R_s^+ = R \cup \{(b, a) \mid (a, b) \in R\}$$
$$= \{(a, b) \mid a \text{ divides } b\} \cup \{(b, a) \mid a \text{ divides } b\}$$
$$= \{(a, b) \mid a \text{ divides } b\} \cup \{(a, b) \mid b \text{ divides } a\}$$
$$= \{(a, b) \mid a \text{ divides } b \text{ or } b \text{ divides } a\}$$

Warshall's Algorithm (Finding the Transitive Closure)

Finding the transitive closure using Warshall's Algorithm

Sometimes it is difficult to find all the ordered pairs in transitive closure of a relation. Warshall's Algorithm is considered an efficient method in finding the transitive closure of a relation.

Example: By using Warshall's algorithm, find the transitive closure of the relation $R = \{(2, 1), (2, 3), (3, 1), (3, 4), (4, 1), (4, 3)\}$ on set $A = \{1, 2, 3, 4\}$.

Solution: First, we will represent the relation R in matrix form.

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}_{4 \times 4}$$

Understand that there are 4 elements in set A . Therefore, 4 steps are required in order to find the transitive closure of relation R [According to Warshall's Algorithm].

- >> In Step 1, we will consider 1st column and 1st row of the above matrix i.e., C_1 and R_1 .

Write all positions where 1 is present in column 1.

$$C_1 = \{2, 3, 4\}$$

Also, write all position where 1 is present in row 1

$$R_1 = \emptyset$$

Now, take the cross product of C_1 and R_1 .

$$C_1 \times R_1 = \emptyset.$$

Therefore, no new additions.

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}_{4 \times 4}$$

- >> In Step 2, we will consider 2nd column and 2nd row of the above matrix.

$$C_2$$

$$R_2$$

$$C_2 \times R_2 = \emptyset$$

$$\emptyset$$

$$\{1, 3\}$$

Therefore, no new additions.

- >> In Step 3, we will consider 3rd column and 3rd row.

$$C_3$$

$$R_3$$

$$C_3 \times R_3 = \{(2, 1), (2, 4), (4, 1), (4, 4)\}$$

$$\{2, 4\}$$

$$\{1, 4\}$$

$$\begin{array}{c}
 1 \quad 2 \quad 3 \quad 4 \\
 \begin{bmatrix}
 0 & 0 & 0 & 0 \\
 1 & 0 & 1 & 1 \\
 1 & 0 & 0 & 1 \\
 1 & 0 & 1 & 1
 \end{bmatrix}
 \end{array}
 \begin{array}{l}
 \text{New Matrix} \\
 4 \times 4
 \end{array}$$

>> In Step 4, we will consider 4th column and 4th row of the above matrix.

$$\begin{array}{cc}
 C_4 & R_4 \\
 \{2, 3, 4\} & \{1, 3, 4\}
 \end{array}$$

$$C_4 \times R_4 = \{(2, 1), (2, 3), (2, 4), (3, 1), (3, 3), (3, 4), (4, 1), (4, 3), (4, 4)\}$$

$$\begin{array}{c}
 1 \quad 2 \quad 3 \quad 4 \\
 \begin{bmatrix}
 0 & 0 & 0 & 0 \\
 1 & 0 & 1 & 1 \\
 1 & 0 & 1 & 1 \\
 1 & 0 & 1 & 1
 \end{bmatrix}
 \end{array}
 \begin{array}{l}
 \text{New Matrix} \\
 4 \times 4
 \end{array}$$

R_t^+

$$R_t^+ = \{(2, 1), (2, 3), (2, 4), (3, 1), (3, 3), (3, 4), (4, 1), (4, 3), (4, 4)\}$$

R_t^+ is the transitive closure of relation R.

Closure of Relations (Solved Problems) (Set 2)

Problem 1: Find the smallest relation containing the relation $\{(1, 2), (1, 4), (3, 3), (4, 1)\}$ defined on a set $A = \{1, 2, 3, 4\}$ that is reflexive and transitive.

Solution:

Reflexive closure of R

$$R_r^+ = R \cup \{(a, a) \mid a \in A\}$$

$$\begin{aligned} R_r^+ &= \{(1, 2), (1, 4), (3, 3), (4, 1)\} \cup \{(1, 1), (2, 2), (3, 3), (4, 4)\} \\ &= \{(1, 1), (1, 2), (1, 4), (2, 2), (3, 3), (4, 1), (4, 4)\} \end{aligned}$$

Transitive closure of R
 $R = \{(1, 2), (1, 4), (3, 3), (4, 1)\}$

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Step 1:

$$\begin{matrix} C_1 & R_1 \\ \{4\} & \{2, 4\} \end{matrix}$$

$$C_1 \times R_1 = \{(4, 2), (4, 4)\}$$

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

Step 2:

$$\begin{matrix} C_2 & R_2 \\ \{1, 4\} & \emptyset \end{matrix}$$

$$C_2 \times R_2 = \emptyset$$

Step 4:

$$\begin{matrix} C_4 & R_4 \\ \{1, 4\} & \{1, 2, 4\} \end{matrix}$$

$$C_4 \times R_4 = \{(1, 1), (1, 2), (1, 4), (4, 1), (4, 2), (4, 4)\}$$

Step 3:

$$\begin{matrix} C_3 & R_3 \\ \{3\} & \{3\} \end{matrix}$$

$$C_3 \times R_3 = \{(3, 3)\}$$

$$R_1^+ = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$R_1^+ = \{(1, 1), (1, 2), (1, 4), (3, 3), (4, 1), (4, 2), (4, 4)\}$$

$$R_t^+ = \{(1, 1), (1, 2), (1, 4), (3, 3), (4, 1), (4, 2), (4, 4)\}$$

$$R_r^+ = \{(1, 1), (1, 2), (1, 4), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_r^+ \cup R_t^+ = \{(1, 1), (1, 2), (1, 4), (2, 2), (3, 3), (4, 1), (4, 2), (4, 4)\}$$

Problem 2: Find the transitive closure of relation $R = \{(a, c), (b, d), (c, a), (d, b), (e, d)\}$ on set $A = \{a, b, c, d, e\}$

Solution:

Let say $a = 1$, $b = 2$, $c = 3$, $d = 4$, and $e = 5$.

$$A = \{1, 2, 3, 4, 5\}$$

$$R = \{(1, 3), (2, 4), (3, 1), (4, 2), (5, 4)\}$$

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Step 1:

C_1

R_1

$$\{3\}$$

$$\{3\}$$

$$C_1 \times R_1 = \{(3, 3)\}$$

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Step 2:

C_2

R_2

$$\{4\}$$

$$\{4\}$$

$$C_2 \times R_2 = \{(4, 4)\}$$

Step 3: C_3 R_3
 $\{1, 3\}$ $\{1, 3\}$

$$C_3 \times R_3 = \{(1, 1), (1, 3), (3, 1), (3, 3)\}$$

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Step 4: C_4 R_4
 $\{2, 4, 5\}$ $\{2, 4\}$

$$C_4 \times R_4 = \{(2, 2), (2, 4), (4, 2), (4, 4), (5, 2), (5, 4)\}$$

$$R_t^+ = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Step 5: C_5 R_5
 $\emptyset$ $\{2, 4\}$

$$C_5 \times R_5 = \emptyset$$

$$R_{\dagger}^+ = \{(1, 1), (1, 3), (2, 2), (2, 4), (3, 1), (3, 3), (4, 2), (4, 4), (5, 2), (5, 4)\}$$

$$R_{\dagger}^+ = \{(a, a), (a, c), (b, b), (b, d), (c, a), (c, c), (d, b), (d, d), (e, b), (e, d)\}$$

Closure of Relations (GATE Problem)

Problem: Consider the binary relation $S = \{(x, y) \mid y = x + 1 \text{ and } x, y \in \{0, 1, 2\}\}$.

The reflexive transitive closure of S is

(A) $\{(x, y) \mid y > x \text{ and } x, y \in \{0, 1, 2\}\}$

(B) $\{(x, y) \mid y \geq x \text{ and } x, y \in \{0, 1, 2\}\}$

(C) $\{(x, y) \mid y < x \text{ and } x, y \in \{0, 1, 2\}\}$

(D) $\{(x, y) \mid y \leq x \text{ and } x, y \in \{0, 1, 2\}\}$

[GATE 2004]

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[GATE 2004]

Solution: $S = \{(x, y) \mid y = x + 1 \text{ and } x, y \in \{0, 1, 2\}\}$
 $y = x + 1$ means $y > x$.

Reflexive closure:

$$\begin{aligned} S_r^+ &= S \cup \{(a, a) \mid a \in \{0, 1, 2\}\} \\ &= \{(x, y) \mid y = x + 1\} \cup \{(x, y) \mid y = x\} \\ &= \{(x, y) \mid y > x\} \cup \{(x, y) \mid y = x\} \\ &= \{(x, y) \mid y \geq x\} \end{aligned}$$

Reflexive Transitive Closure:

$$\begin{aligned} S_r^+ \cup S_r^+ &= \{(x, y) \mid y \geq x\} \cup \{(x, y) \mid y > x\} \\ &= \{(x, y) \mid y \geq x\} \\ &= \{(x, y) \mid y \geq x \text{ and } x, y \in \{0, 1, 2\}\} \end{aligned}$$

Transitive closure:

$$\begin{aligned} S_t^+ &= S \cup \{(x, z) \mid (x, y) \in S \wedge (y, z) \in S\} \\ &= \{(x, y) \mid y = x + 1\} \cup \\ &\quad \{(x, z) \mid y = x + 1 \wedge z = y + 1\} \\ &= \{(x, y) \mid y > x\} \cup \{(x, z) \mid y > x \\ &\quad \wedge z > y\} \\ &= \{(x, y) \mid y > x\} \cup \{(x, z) \mid z > x\} \\ &= \{(x, y) \mid y > x\} \end{aligned}$$

Therefore, option (B) is the correct option.

Equivalence Relation

Definition: A relation R on a set A is an equivalence relation iff R is reflexive, symmetric, and transitive.

Example: $A = \{0, 1, 2, 3\}$
 $R_1 = \{(0, 0), (1, 1), (2, 2), (3, 3)\}$
Is R_1 an equivalence relation?
Yes.

$R_2 = \{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$
Is R_2 an equivalence relation?
Is R_2 reflexive?
No. Because $(1, 1)$ is not a member of R_2 .

$R_3 = \{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$
Is R_3 an equivalence relation? Ask yourself: Is R_3 reflexive? Yes
Is R_3 symmetric? Yes
Is R_3 transitive? Yes

$$R_4 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$$

Is R_4 an equivalence relation? Ask yourself: Is R_4 reflexive? Yes
Is R_4 symmetric? No.

Therefore, R_4 is not an equivalence relation.

$R_5 = \emptyset$ Is R_5 an equivalence relation?
Ask yourself: Is R_5 reflexive? No
Is R_5 symmetric?

Therefore, R_5 is not an equivalence relation.

$R_6 = AXA$ Is R_6 an equivalence relation?
Ask yourself: Is R_6 reflexive? Yes
Is R_6 symmetric? Yes
Is R_6 transitive? Yes

Therefore, R_6 is an equivalence relation.